

Assingment - 2

CSE - IIII

```
1) #include <stdio.h>
    #include <math.h>
```

```
int count_divisors (int n) {
```

```
    int count = 0;
```

```
    for (int i = 1; i <= n; i++) {
```

```
        if (n % i == 0) {
```

```
            count++;
```

```
        }
```

```
    return count;
```

```
int is_perfect_square (int n) {
```

```
    int root = sqrt(n);
```

```
    return (root == n) ? 1 : 0;
```

```
main ( )
```

Σ

```
int n, m;
```

```
scanf("%d %d", &n, &m);
```

```
int divisor_n = count_divisor(n);
```

```
int divisor_m = count_divisor(m);
```

```
int p_sq_n = is_perfect_square(n);
```

```
int p_sq_m = is_perfect_square(m);
```

```
if (divisor_n > divisor_m) printf("1st  
number has more divisor");
```

```
else printf("2nd number has more divisor");
```

```
if (p_sq_n) printf("1st is perfect square");
```

```
else printf("no");
```

```
if (p_sq_m) printf("2nd is perfect square");
```

```
else printf("no");
```

```
return 0;
```

b)

$$x = 12, y = 80, a = 12, b = 12$$

$$12 * 12 + 12 = 156$$

$$a = 232, b = 7, x = 12$$

2 a

ite	i	j	S1[]	S2[]
0	0	0	S1[0] = A	E
	1	1	S1[1] = B	F
	2	2	S1[2] = C	G
	3	3	S1[3] = d	H
	4	4	S1[4] = 7	7
	5	5	S1[5] = 8	8
	6	6	S1[6] = 9	9
	7	7	S1[7] = 10	

output : S1 = ABCd789

(b) #include <stdio.h>

int main()

{

char str[100]; dest[100];

gets(str);

myStrRev(str, dest);

return 0;

}

void myStrRev(char src[], char dest[]) {

~~int i = 0;~~

int len = 0;

while (*src) {

len++;

src++;

}

~~for (int j = 0; j < len; j++) {~~

~~dest[j] =~~

for (int i = 0, j = len - 1; i < len; i++, j--)

{ dest[i] = src[j];

}

}

③ a. i) comp

ii)

iii) compute

b. 672454440

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c. #include <stdio.h>

```
int factorial(int n) {
```

```
    if (n == 1) return 1;
```

```
    return factorial(n-1) * n;
```

```
}
```

```
int main() {
```

```
    int n;
```

```
    scanf("%d", &n);
```

```
    factorial printf("%d/n", factorial(n));
```

```
    return 0;
```

```
}
```